

How to load data

Dealing with PDFs

- PDFs are terrible files for storing data
- Some PDFs are machine-readable, others aren't
 - Machine-readable means the underlying data in the PDF is still intact
 - How can you tell? Try to highlight the text.

Dealing with PDFs

- If you can highlight the data in the PDF, it **IS** machine-readable
 - Try an online tool like Tabula to get the data into a spreadsheet
- If you can't highlight it, i.e. if it looks like a scanned document or a photo, it's **NOT** machine-readable
 - You'll need OCR (optical character recognition software) to convert into machine-readable data. Try Acrobat or Document Cloud or new Excel hotness
 - OR go back to your data source and ask for a machine-readable file

→ Coming up: Tabula

`tabula.technology`

Tabula is a free, open-source tool that extracts tables from machine-readable PDFs into a spreadsheet.

We'll use it hands-on in a later session — but keep it in mind any time a source hands you a PDF full of data.

File Types

- Data can be stored in lots of file types
- Text files - the simplest. Basically a text document of data
- File type is usually .txt
 - Obviously a 10,000-page document of data isn't useful. We have to separate it somehow.
 - Generally two ways to do this

File Types

- Check the file name extension (.csv, .txt, etc.)
- Open the file up in a text editor and browse
- Do you have a header row?
- What's separating each field?

dc311sample.csv

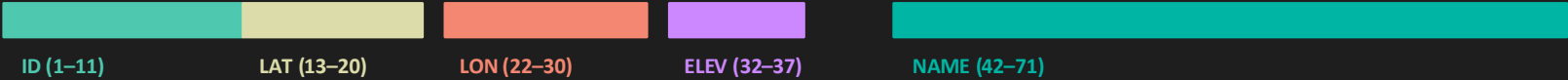
```
X,Y,SERVICEREQUESTID,STREETADDRESS,  
CITY,STATE,ZIPCODE,SERVICECODE,  
SERVICECODEDESCRIPTION,...,  
ADDDATE,RESOLUTIONDATE,...,  
WARD,LATITUDE,LONGITUDE,...
```

```
26-00183801,5224 AMES ST NE,  
WASHINGTON,DC,20019  
2026/03/25 04:00:18+00,  
2026/03/30 14:39:05+00,  
Ward 7
```

ghcnd-stations.txt - ncei.noaa.gov/pub/data/ghcn/daily/ghcnd-stations.txt

1 2 3 4 5 6 7 8
12345678901234567890123456789012345678901234567890123456789012345

ACW00011604	17.1167	-61.7833	10.1	ST JOHNS COOLIDGE FLD		
AE000041196	25.3330	55.5170	34.0	SHARJAH INTER. AIRP	GSN	41196
AFM00040948	34.5660	69.2120	1791.3	KABUL INTL		40948
AG000060390	36.7167	3.2500	24.0	ALGER-DAR EL BEIDA	GSN	60390
ALM00013615	41.4150	19.7210	38.4	TIRANA RINAS	GSN	13615



No header row. No delimiters. You must know the column positions from the schema.

File Type: Fixed Width

- You must manually put breaks in the data to tell your computer where a new field ends and another begins
- If you have a fixed width file, make sure your source gives you a guide or schema that tells you where to put the breaks. Don't guess!
 - Example

File Type: Delimited

- The data fields are separated by a delimiter (often punctuation) that tells our computer where a new field ends and another begins
 - This is more common than fixed width
 - Common delimiters
 - Commas
 - Tabs
 - Pipes (|)

File Type: CSV

- Commas are the most common delimiters
- Delimited files that use commas are also known as CSVs, aka comma-separated values
- File type can be .csv instead of .txt
- **But what if you have commas in a field?**
 - A text qualifier (usually quote marks) tells the computer to ignore commas within fields
 - Example: "Trash Cart Repair, urgent" will look like "Trash Cart Repair, urgent" in the text file

```
dc311sample.csv (in text editor)
```

```
SERVICEREQUESTID,STREETADDRESS,SERVICECODEDESCRIPTION
```

```
26-00183801,5224 AMES ST NE,  
Supercan - Delivery
```

```
26-00183803,4528 28TH STREET NW,  
"Trash Cart Repair,  
urgent"
```

```
↑ comma delimiter  
↑ text qualifier (quotes)
```

dc311sample.csv – opened in text editor

```
SERVICEREQUESTID,STREETADDRESS,CITY,STATE,ZIPCODE,SERVICECODE,SERVICECODEDESCRIPTION,...,ADDDATE,RESOLUTIONDATE,WARD
```

```
26-00183801,5224 AMES STREET NE,WASHINGTON,DC,20019,S0422,Supercan - Delivery,...,2026/03/25 04:00:18+00,2026/03/30 14:39:05+00,Ward 7
```

```
26-00183802,43 ANACOSTIA ROAD NE,WASHINGTON,DC,20019,S0336,Out of State Parking Violation (ROSA),...,2026/03/25 04:01:35+00,2026/03/25 06:20:07+00,Ward 7
```

```
26-00183803,4528 28TH STREET NW,WASHINGTON,DC,20008,TRACO001,Trash Cart Repair,...,2026/03/25 04:16:01+00,2026/04/07 20:12:02+00,Ward 3
```

```
26-00183804,1625 ECKINGTON PL NE,WASHINGTON,DC,20002,S0311,Rodent Inspection and Treatment,...,2026/03/25 04:16:01+00,2026/03/30 14:17:50+00,Ward 5
```

↑ header row

← comma delimiter separates each field

data FIELD types

Just like there are different types of data files, there are different types of data fields (aka columns aka variables).

How you upload them can make a huge difference.

data FIELD types

Text / character

- Alphanumeric
- Short or long
- You don't need to do calculations with these things
- Ex in 311: SERVICEREQUESTID, STREETADDRESS, SERVICECODEDESCRIPTION, WARD

Numeric

- Things you want to calculate with
- Ex: counts, sums, dollar amounts
- Can be integers or decimal numbers
- Ex in 311: SERVICECALLCOUNT, LATITUDE, LONGITUDE

Dates / times

- Ex: 2026/03/25 04:00:18+00
- Storing these as dates instead of text lets you calculate days between open and close
- Ex in 311: ADDDATE, RESOLUTIONDATE, SERVICEDUEDATE

Why should you care about types of data fields?

- **Different field types sort differently!!**
- If you're not careful when you load your data, Excel/Google Sheets will try and be helpful and guess your data types.
 - It's usually wrong.
 - **THUS → DO NOT JUST DOUBLE CLICK YOUR DATA FILE TO OPEN IT!!!!**

Computers are dumb

How does Excel/Google Sheets get things wrong when guessing field types?

- **Drops leading zeros**
 - DC zip codes like 20001, 20036 are fine — but some jurisdictions use codes starting with 0
 - Excel will assume ZIPCODE is a number and strip leading zeros
 - Avoid this by explicitly loading as a text field
- **Reformat numbers to dates**
 - 311 dates look like: 2026/03/25 04:00:18+00
 - Excel may misread the format and scramble these — or treat them as text strings you can't calculate with
 - Avoid this by explicitly loading as a date field

Watch out in 311 data

ZIPCODE — load as text, not number

SERVICEREQUESTID (e.g. 26-00183801) — text

WARD — load as text (don't average wards!)

ADDDATE / RESOLUTIONDATE

— explicitly load as dates so you can calculate days to resolution